Indian Institute of Technology Palakkad

Department of Mechanical Engineering

ME 5610: Soft Computing (**Part A**)

End-semester exam (06 May 2025)

Time: 9:30 AM - 12:00 PM **Marks: 25**

Note to invigilators and students :

(i) **Please note that the full duration of the exam is 2 hours 30 minutes.**

(ii) **The full question paper containing Part A and Part B should be given to students at the beginning of the examination.**

(iii) **Separate answer booklets should be given for Part A and Part B.**

(iv) **Both the answer booklets (Part A and Part B) should be collected at the end of the examination.**

Instructions to students:

(i) **Answer all questions. Questions carry marks as stated against them**

(ii) **Assume any missing data suitably**

Q1. There are two electrical generators G_1 and G_2 whose power outputs are P_1 and P_2 MW, respectively. The generators are connected to a load line such that

$$P_1 + P_2 = 800$$

The cost of producing power from the generator is given by following equations

$$C_1 = 300 + 7.3P_1 + 0.0034(P_1)^2$$
$$C_2 = 200 + 8P_2 + 0.0019(P_2)^2$$

Determine P_1 and P_2 using the method of Lagrange multipliers such that the total cost is minimum. Furthermore, determine the value of the Lagrange multiplier (λ) and comment on its significance. **[5]**

Q2. Determine whether the given direction ($\mathbf{d}$) at the point ($\mathbf{x}$) is that of descent for the following function (show all of the calculations) **[3]**

$$f(\mathbf{x}) = 9 - 8x_1 - 6x_2 - 4x_3 - 2x_1^2 + 2x_2^2 + x_3^2 + 2x_1x_2 + 2x_2x_3;$$
$$\mathbf{d} = (-2, -2, 0);$$
$$\mathbf{x} = (1, 1, 1)$$

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Q3. An experiment was designed to examine the effect of load, x (in appropriate units) on the probability of failure of specimens of a certain industrial component. The following results were obtained:

Load, x	Number of specimens	Number of failures
10	400	20
30	500	50
60	450	110
85	600	275
95	550	300
110	350	250

The regression model suitable for this problem is of the following form:

$$p = \frac{ax}{(b+x)}$$

where p is the probability of failure of the component.

(a) Linearise the model and obtain the "best" estimates of a and b. [6]

(b) Also obtain the R^2 associated with the "best" fit model. [2]

Q4. It is required to minimize the following function

$$f(\mathbf{x}) = 25x_1^2 + 20x_2^2 - 2x_1 - x_2; (\mathbf{x} \in \mathbf{R})$$

Starting from an initial guess of (3,1), perform one iteration of the steepest-descent algorithm and compare the values of the objective function at the guess point and the new point obtained after one iteration. [5]

Q5. Write whether True/False. Also give an associated brief explanation. [4]

(i) The steepest-descent method can converge to a local maximum point starting from a point where the gradient of the function is non-zero.

(ii) In unconstrained optimization, the slope of the cost function along the descent direction at zero step size is always positive.

(iii) In unconstrained optimization, a direction of descent can be found at a point where the gradient of the cost function is zero.

(iv) In an optimization search algorithm, the step size determination is always a one-dimensional problem.

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END SEMESTER EXAMINATION

(Part - B)

SOFT COMPUTING (ME5610)

Duration (Part A+ Part B): 2 hours 30 minutes **Maximum marks: 25**

Note:

- **Answer all the questions.**
- **Assume suitable missing data.**
- **Any doubts related to the question paper will not be addressed by the invigilators/ course instructor(s) during the examination.**

1. Using Genetic Algorithm, maximize the function $f(x) = x^2$ with x intervals [10 to 30]. Consider the given initial population 14, 17, 24, 22.

   Note: Consider the initial population as the first generation; stop the iterations after calculating two more subsequent generations.

   [10 Marks]

2. Mention the mathematical and graphical representations of the following transfer functions used in artificial neural networks.

   a) Hard limit transfer function

   b) Linear transfer function

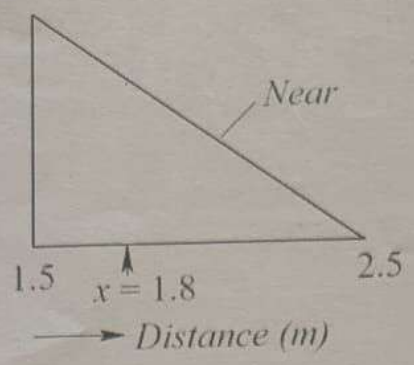
   (No need to explain the activation functions.. Just mention the mathematical and graphical representations).

   [4 Marks]

3. Let A(x) and B(x) are two fuzzy sets, A(x) = {(x_1, 0.2), (x_2, 0.3), (x_3, 0.4), (x_4, 0.5)}; B (x) = {(x_1, 0.4), (x_2, 0.5), (x_3, 0.6), (x_4, 0.7)}. Determine inaccuracy of the fuzzy set B (x) with respect to the fuzzy set A (x).

   [3 Marks]

4. Graph shown below represents a triangular membership function for the distance with the linguistic variable 'Near'. Find the membership function value of x at distance 1.8 m.



[1 Mark]

5. Mention the mathematical and graphical representations of the trapezoidal membership function. *(No need to explain the activation functions.. Just mention the mathematical and graphical representations).* **[3 Marks]**

6. If A(x) and B(x) are two fuzzy sets.

   A(x) = $\{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3), (x_4, 0.4)\}$

   B(x) = $\{(x_1, 0.4), (x_2, 0.6), (x_3, 0.9), (x_4, 0.8)\}$

   (a) Find the algebraic sum of the two fuzzy sets.

   (b) Find the bounded sum of the two fuzzy sets.

[4 Marks]

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